

Blackbird State

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Abstract

Your abstract.

1 Introduction

The blackbird bipedal needs to store its state information in a gym environment. here is the state that is defined

2 State

2.1 Bipedal State

1. Torso Positions (x, y, z)
2. Torso Angles (q_1, q_2, q_3, w) // they are represented as Quaternions
3. Torso Linear Velocity ($\dot{x}, \dot{y}, \dot{z}$)
4. Torso Angular Velocity ($\dot{\rho}, \dot{\phi}, \dot{\gamma}$)
5. Joint Positions ($\theta_1 \dots \theta_{10}$)
6. Joint Velocities ($\dot{\theta}_1 \dots \dot{\theta}_{10}$)
7. Contact States (Torso, footL, footR)

This gives us a total of $6(position/orientation) + 6(velocities) + 20(jointstates) + 3(contactsensor) = 35$ dimensions in our state vector

2.2 How to Incorporate this state

We have altogether 35 different states to observe. below, we list each

2.3 Bipedal State

1. Pose.X: 0
2. Pose.Y: 1
3. Pose.Z: 2
4. ρ : 3
5. ϕ : 4
6. γ : 5
7. ($\dot{x}, \dot{y}, \dot{z}$)
8. Torso Angular Velocity ($\dot{\rho}, \dot{\phi}, \dot{\gamma}$)

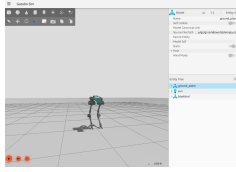


Figure 1: The star of the show

9. Joint Positions ($\theta_1 \dots \theta_{10}$)
10. Joint Velocities ($\dot{\theta}_1 \dots \dot{\theta}_{10}$)
11. Contact States (Torso, footL, footR)

References